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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☐ ☒ A description of all covariates tested
- ☐ ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐ ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Data was downloaded and further curated using custom code written in R (version 4.1.1). We used a custom approach for initial data filtering and preprocessing as described in more detail in the Methods. Custom code is available on Github in https://github.com/tiroshlab/3ca/tree/main/ITH_hallmarks
Data analysis	We used open source code and software for some of the analysis: we used SCENIC version 1.3.1 for inferring gene regulators, inferCNA for CNA analysis (https://github.com/jlaffy/infercna), Cytoscape version 3.9.1 for graph visualization, the 'NMF' R package (https://cran.r-project.org/web/packages/NMF/index.html) and QuPath version 0.3.2 for analyzing IHC slide images. We used custom written code using R version 4.1.1 for multiple analyses, including standard QC steps, cell annotations, clustering, visualization, MP-generation. Relevant code can be found on github in https://github.com/tiroshlab/3ca/tree/main/ITH_hallmarks

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

All curated data is available at (<https://www.weizmann.ac.il/sites/3CA>), aside from samples from unpublished studies that will be added when possible or cases in which sharing permission from the authors was denied.

Human research participants

Policy information about [studies involving human research participants and Sex and Gender in Research](#).

Reporting on sex and gender	We used mainly published data, as described above, which included samples from males and females. To our knowledge, the data was not deliberately enriched for a particular sex in none of the studies, and other than specific cancer types which are sex specific (e.g. breast cancer or prostate cancer), the data was evenly distributed. We included any available metadata, including sex, for each sample, which is provided in the 3CA website.
Population characteristics	Similar to sex, we made major efforts to include other sample information, including age, treatment status and more. All collected information can be found on the 3CA website.
Recruitment	We prioritized high quality and available data for our analysis. Although we put major efforts to obtain data from a variety of tumor types, there are some types that are under-represented in our compendium (e.g. endometrial cancer or gastric cancer). Hence, we will be able to generalize our findings to these types after accessing the relevant data (work in progress).
Ethics oversight	Ethics protocols were typically obtained by the original authors. For unpublished data that was sequenced by our lab or collaborators we used....For the sample sections used for the IHC staining experiments we used....

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	We curated 1163 samples from 77 studies. Following QC, we typically used all available samples within each study. Depending on the question we tried to address, analyses were performed within each study, across samples from the same cancer type, or across samples that participated in each MP.
Data exclusions	We excluded several samples in which the malignant cell annotations were questionable, based on pre-established criteria for CNA signal and correlation, as described in the Methods. After generating MPs, we also excluded a few programs that were enriched with low quality genes (ribosomal or mitochondrial).
Replication	We believe that our findings are highly reproducible. First, our definitions for MPs were derived using a very large dataset that was sequenced by dozens of different groups using different methods. We used standard IHC staining in multiple samples of different cancer types to validate some of our findings, and also further validated our findings using published CODEX and VISIUM data, and TCGA data.
Randomization	Randomization was not relevant to our study since we aimed to infer ITH in an unsupervised manner based on a large compendium of available published data. We prioritized studies which utilized high quality data according to our judgment and that sequenced many samples. We put special effort to include also rarer cancer types, which resulted in 24 cancer types altogether.
Blinding	Blinding was not possible or relevant in this study since we did not perform a clinical trial or any other experiment/analysis that might require blinding, but rather curated all scRNAseq data that was published as part of other studies and available to us.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

We used antibodies for our IHC validation experiments (Table S15):

- mouse anti-human MHC class II; Abcam; ab55152; donkey anti-rabbit Cy3; Jackson; 711-165-152
- rabbit anti-human ISG20; Abcam; ab154393; donkey anti-goat AlexaFluor 647; Jackson; 705-605-003
- goat anti-human SMA; LSBio; LS-b3933-50; donkey anti-rat AlexaFluor 647; Jackson; 712-605-150
- goat anti-human p63; R&D; AF1916; donkey anti-goat Cy3; Jackson; 705-165-003
- rat anti-human CD3; LSBio; LS-B8765-50; donkey anti-rabbit FITC; Jackson; 711-095-152
- goat anti-human EPCAM; Millipore; AF960; donkey anti-mouse FITC; Jackson; 715-095-150
- rabbit anti-human TFF1; Abcam; ab92377
- mouse anti-human pan-cytokeratin; Abcam; ab86734
- rat anti-human MECA79; NovusBiologicals; NB100-77673
- rabbit anti-human C3; Abcam; ab97462
- rabbit anti-human SLPI; Invitrogen; PA582990
- rabbit anti-human SPRR1B; LSBio; LS-C161474-400
- mouse anti-human LAMC2; Novus Biologicals; NBP2-42388

Validation

Below we include respective information on antibody, cat. number or clone, manufacturer, species, expected expression and validation information:

- 1) Pan -Cytokeratin; ab234297; Abcam; rabbit; epithelial, cancer; tested in human skin tissue by Abcam, tested in tumor-adjacent normal epithelium in HNSCC by us
- 2) EPCAM; AF960; Millipore; goat; epithelial, cancer; tested in PDAC, HNSCC, and tumor-adjacent normal epithelium by us
- 3) p63; AF1916; R&D; goat; squamous, cancer; tested in human breast by R&D, tested in HNSCC by us
- 4) SLPI; PA582990; Invitrogen; rabbit; epithelial, cancer; tested in human cervix, uterine, and skeletal muscle by Invitrogen, tested in HNSCC by us
- 5) SPRR1B; LS-C161474-400; LSBio; rabbit; epithelial, cancer; tested in human breast carcinoma by LSBio, tested in HNSCC - by us
- 6) ISG20; ab154393; Abcam; rabbit; immune; tested in non-Hodgkin's lymphoma and tumor xenograft by Abcam, tested by us in human lymph node
- 7) CD31; ab9498; Abcam; mouse; endothelial; tested in human tonsil and lung by Abcam, tested in human placenta by us
- 8) C3; ab97462; Abcam; rabbit; immune tested in mouse brain by Abcam, tested in human lymph node and tonsil by us
- 9) CD3; LS-B8765-50; LSBio; rat; T cell; tested in human thymus by LSBio, tested in human lymph node, tonsil, and spleen by us
- 10) aSMA; LS-b3933-50; LSBio; goat; myofibroblast/CAF; tested by us in human spleen and placenta
- 11) TFF1/Anti-Estrogen Inducible Protein pS2; ab92377; Abcam; rabbit; mucosal glands/goblet cells, cancer; tested in human breast and ovarian carcinoma by Abcam, tested in PDAC by us
- 12) MHC class II; ab55152; Abcam; mouse; immune; tested in human bowel tissue by Abcam, tested in human lymph node by us
- 13) MECA79; NB100-77673; Novus Biologicals; rat; high endothelial venules (HEVs); tested in lymph node and tonsil by Novus, tested in human tonsil by us
- 14) LAMC2; NBP2-42388; Novus Biologicals; mouse; pEMT (cancer), ECM; tested in fallopian tube and liver by Novus, HNSCC in Puram et al. 2017 <https://doi.org/10.1016/j.cell.2017.10.044>

Eukaryotic cell lines

Policy information about [cell lines and Sex and Gender in Research](#)

Cell line source(s)

State the source of each cell line used and the sex of all primary cell lines and cells derived from human participants or vertebrate models.

Authentication

Describe the authentication procedures for each cell line used OR declare that none of the cell lines used were authenticated.

Mycoplasma contamination

Confirm that all cell lines tested negative for mycoplasma contamination OR describe the results of the testing for mycoplasma contamination OR declare that the cell lines were not tested for mycoplasma contamination.

Commonly misidentified lines
(See [ICLAC](#) register)

Name any commonly misidentified cell lines used in the study and provide a rationale for their use.

Animals and other research organisms

Policy information about [studies involving animals](#); [ARRIVE guidelines](#) recommended for reporting animal research, and [Sex and Gender in Research](#)

Laboratory animals

We used a few published mouse and PDX models, as described in Table S1: Ebinger et al 2016 (ALL PDX models), Ireland et al 2020 (SCLC mouse model), Yao et al 2020 (ESCC mouse model), and unpublished GBM mouse model

Wild animals

The study did not involve wild animals.

Reporting on sex

We used mainly published datasets, as described above, which included samples from males and females. To our knowledge, the data was not deliberately enriched for any sex in none of the studies, and other than specific cancer types which are sex specific (e.g. breast cancer or prostate cancer), the data was evenly distributed. We included any available metadata, including gender, for each sample, which is provided in the 3CA website.

Field-collected samples

This study did not involve samples collected from the field.

Ethics oversight

In most cases we used published data that did not require further ethical approval.
For unpublished data that was sequenced by our lab or collaborators we used...
For samples used for IHC staining we used....

Note that full information on the approval of the study protocol must also be provided in the manuscript.